CAMBRIDGE INSTITUTE OF TECHNOLOGY
K R PURAM, BANGALORE-560036
Department of Computer Science & Engineering
Final Year B.E Project work Phase-1 : 2025-26

Synopsis

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Title: End-to-End Brain Tumor Detection using Deep Learning (PyTorch & Flask)

Brain tumors are caused by abnormal and uncontrolled cell growth in the brain and are among the most serious neurological disorders. Early and accurate detection using Magnetic Resonance Imaging (MRI) is essential for effective treatment and improving patient survival. However, manual analysis of MRI scans by radiologists is often complex, time-consuming, and prone to variability. Therefore, automated computer-aided diagnostic systems based on Artificial Intelligence are increasingly used in medical image analysis.

This project proposes an end-to-end brain tumor detection and classification system using Deep Learning and Machine Learning techniques. Convolutional Neural Networks (CNNs) with transfer learning models such as ResNet are used to extract deep features from MRI images, which are then classified using Support Vector Machine (SVM) for improved accuracy. The system performs multiclass classification of brain tumors including Glioma, Meningioma, Pituitary tumor, and No Tumor. Image preprocessing and data augmentation techniques such as normalization, resizing, rotation, and flipping are applied to enhance model performance.

The trained model is deployed through a Flask-based web application where users can upload MRI images and receive real-time predictions. Explainable Artificial Intelligence techniques such as Grad-CAM are also used to highlight tumor-affected regions, improving the interpretability of the results. The proposed system aims to provide an efficient AI-based tool that assists healthcare professionals in early brain tumor detection and clinical decision-making.

Sustainable Development Goals (SDG)

1. Good Health and Well-Being.

Name of the Guide: Mr.Lokesh **Signature of Guide:** **Date:**

Signature of Project coordinator: